

CHANDAN KUMAR N S

International Institute of Information Technology, Bangalore

+91-8495008958 ✉ chandankumar.ns@iiitb.ac.in [linkedin.com/in/chandan-kumar-n-s-8578b521b/](https://www.linkedin.com/in/chandan-kumar-n-s-8578b521b/) github.com/dhanu-07

Education

International Institute of Information Technology, Bangalore <i>Master of Science by Research (VLSI)</i>	2024 – Present 3.51/4 CGPA
International Institute of Information Technology, Bangalore <i>PG Diploma in Advanced VLSI</i>	2024 3.52/4 CGPA
Government SKSJ Technological Institute <i>Bachelor of Engineering in Electronics and Communication</i>	2020 – 2024 8.42/10 CGPA

Internship Experience

Central Manufacturing Technological Institute <i>Sensor Technology Development Centre Intern</i>	Aug 2023 – Sep 2023 Bangalore
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- Developed Ultrasonic and Accelerometer Sensors for industrial applications.

Research & Publications

Hardware Efficient Multiplier design using Approximate Booth Encodings

- Accepted in **IEEE International Conference on Computer Design (ICCD), 2025** (*Selected as a Best Paper Nominee*).
- Developed a novel methodology for encoder-level approximation in Radix-4 Booth multipliers to reduce hardware complexity by eliminating specific control signals.
- Utilized NSGA-II to identify optimal application-specific configurations for JPEG compression, MLP inference, and μ -law processing.

BM-HPR: Error Tolerant Booth Modified Redundant Multiplier

- Accepted in **Intl. Symposium on Quality Electronic Design (ISQED), 2026**.
- Architected a fault-tolerant multiplier for 16-bit datapaths by integrating Radix-4 Booth encoding into an HPR framework, eliminating the tunable K parameter.
- Achieved 65.1% power reduction and 31.8% delay improvement compared to standard TMR.

Hardware Software Co-Optimization for RISC-V HDC Architectures

- Accepted in **Intl. Symposium on Quality Electronic Design (ISQED), 2026**.
- Proposed a hardware-optimized RISC-V co-processor architecture for Hyperdimensional Computing (HDC) using custom SIMD instructions.
- Boosted throughput by 21.57x and verified dimensionality pruning via genetic algorithms on FPGA and ASIC platforms.

Compact-Precision FP8 Multiplier for Diverse AI Workloads

- Submitted to **Great Lakes Symposium on VLSI (GLSVLSI), 2026**.
- Designed FP8 multipliers using reduced-precision internal cores (FP6/FP4) while maintaining E_4M_3 I/O compliance.
- Achieved 71% power savings over standard FP8 baselines with robust accuracy in ResNet and BERT models.

ApproxMUL: Evolutionary Optimization of Error-Resilient Multipliers

- Submitted to **IEEE Annual Symposium on VLSI (ISVLSI), 2026**.
- Introduced a multi-objective evolutionary framework (NSGA-II) for generating unsigned 8-bit multipliers using approximate 3:2, 4:2, and 5:2 compressors.
- Reduced Power-Area-Delay Product (PADP) by 97% with minimal accuracy degradation in CNN workloads.

Technical Skills

Languages: Verilog, Python, C

EDA Tools: OpenRoad, Yosys, Xilinx Vivado, Cadence Virtuoso, OpenLane, Altium, Keil uVision 5, Matlab

Methodologies: Approximate Computing, Multi-Objective Optimization (NSGA-II), ASIC Design Flow

Relevant Coursework

- System Design Through FPGA
- Digital CMOS VLSI Design
- Physical Design Using ASIC
- Machine Learning
- VLSI Architecture Design

Certifications

- VSD-SoC Design of the PicoRV32 RISC-V micro-processor, Udemy
- VSD-Static Timing Analysis I & II, Udemy
- Basics of Machine Learning, Great Learning
- Samsung Chip Design Studio Program, IIIT-Bangalore

Achievements/Extra-curricular

- Finalist - Hack4Sustainability, Siemens and Smile Foundation
- Finalist - PitchPerfect, BMS College of Engineering, Bangalore
- Project Exhibition and Paper Presentation - 18th Karnataka State Level ISTE Student Convention
- Core Member of various committees at Government SKSJ Technological Institute